

ZEROED IN

Data Portfolio Interview Guide

Projects 1–4: what the work demonstrates, how to inspect it, and how to explain it naturally.

Prepared for Rhayko | September 2026

Four cases, one complete analyst workflow

These projects use synthetic data. They demonstrate methods and decision-making; they do not claim real organizational or physiological findings.

Project	Domain	Primary proof
1. Operations KPI Dashboard	Operations / Manufacturing	Excel formulas, PivotTable, KPI logic, data checks
2. Business & Sales Analysis	Business / Sales	SQL grain, joins, CTEs, windows, contribution bridge
3. Workforce Planning Analysis	Workforce / HR	Pandas, missing data, confidence interval, group comparison
4. Human Performance Research	Research	Paired design, correlation, regression, Matplotlib

The story to tell

I can start with raw records, validate the grain and definitions, clean only what can be justified, calculate traceable metrics, present the decision clearly, and document what the data cannot establish.

How to prepare

- Open each README before opening code.
- Trace one headline number back to its source rows.
- Practice explaining one design choice and one limitation for every project.
- Never present a simulated result as a result from an employer, client or study.

Operations KPI Dashboard

Business question: where should a fictional fulfillment operation investigate missed dispatch cutoffs while watching quality and labor usage?

TOOLS	Excel; formulas; SUMIFS/COUNTIFS; exact lookup logic; PivotTable; dashboard charts; CSV; validation scripts.
WORKFLOW	Generate 528 date/site/shift records → validate accounting relationships → load the Excel source table → aggregate selected records → calculate weighted KPIs → compare formula output with a PivotTable and independent checks.
FILES TO OPEN	Operations-KPI-Dashboard-v1.1.xlsx; README.md; shift_operations.csv; data-dictionary.csv; release-checks.json.

What the results support

The designed sample contains 124,665 orders and a 92.5% on-time dispatch rate. South PM is the clearest investigation target. The analysis supports a workload and process review, not a causal staffing claim.

Assumptions and limits

All data are synthetic. Late means after a dispatch cutoff, not late customer delivery. Backlog, cancellations and cross-day carryover are outside scope. KPI targets are illustrative.

How to describe it to HR or a recruiter

"I built an Excel operations dashboard that lets a manager filter site and shift, see service, rework, overtime and throughput together, and trace every KPI back to the source table."

How to describe it to a technical interviewer

"I calculate rates from matching numerators and denominators instead of averaging row percentages. SUMIFS drives the filtered dashboard, an exact lookup supports labeled definitions, and an independent PivotTable plus release checks verifies the totals."

Business & Sales Analysis

Business question: which sales channel needs a margin review before the next promotional budget is set?

TOOLS	SQLite; SELECT/WHERE; INNER and LEFT JOINS; GROUP BY; CASE; CTEs; subqueries; LAG window function; Power BI measures and model notes.
WORKFLOW	Generate six related tables → enforce keys and constraints → aggregate returns to line grain → aggregate lines to order grain → join channel-month spend once → calculate contribution → audit conservation and demonstrate join inflation.
FILES TO OPEN	analysis.sql; model.sql; commerce.sqlite; README.md; POWER-BI.md; channel_month_economics.csv; test_model.py.

What the results support

The simulation produces \$218,018.90 in net sales and \$22,414.06 in contribution after marketing. Paid social is negative in the designed data because deeper discounts, higher returns and marketing costs were encoded into the generator.

Assumptions and limits

Contribution is not operating profit. Fixed overhead, salaries, taxes and financing are excluded. Channel relationships are simulation assumptions, not causal findings. The Power BI material is a documented handoff, not a claimed native PBIX build.

How to describe it to HR or a recruiter

"I built a SQL sales case that goes beyond revenue and shows how returns, product cost, fulfillment and marketing change channel contribution."

How to describe it to a technical interviewer

"The key technical issue is grain. I aggregate returns before joining lines and aggregate orders before joining monthly spend, so one-to-many joins cannot multiply revenue or cost. The analysis also shows CTEs, CASE logic, subqueries and LAG."

Workforce Planning Analysis

Business question: where should HR ask better questions about voluntary turnover, workload and engagement?

TOOLS	Python; Pandas; NumPy; missing-data handling; group summaries; Wilson confidence interval; Welch statistic; Matplotlib; automated tests.
WORKFLOW	Generate 720 fictional employees → validate columns, keys and exit coding → retain 29 missing engagement scores → aggregate by department → calculate turnover and uncertainty → compare overtime groups → export chart and evidence files.
FILES TO OPEN	README.md; analyze.py; workforce_records.csv; department_summary.csv; report.json; test_analysis.py.

What the results support

The simulated turnover rate is 13.8% with a 95% Wilson interval of 11.4%–16.5%. Employees who exited average 6.9 more Q4 overtime hours. That is a department-level investigation signal in the simulation, not an individual risk score.

Assumptions and limits

The cross-sectional design omits compensation, manager practices and labor-market context. The large-sample p-value is approximate. No result supports an employment action or a causal claim.

How to describe it to HR or a recruiter

“I built a workforce case that handles missing survey data honestly and turns employee records into department-level questions without pretending the analysis should decide who is at risk.”

How to describe it to a technical interviewer

“I keep missing engagement values as null, calculate rates from counts, report a Wilson interval for the overall proportion and use a Welch comparison because the two overtime groups have different sizes and may have different variances.”

Human Performance Research

Research question: in a simulated crossover study, how do rested and restricted conditions differ in reaction time, and how is sleep associated with reaction time?

TOOLS	Python; Pandas; NumPy; descriptive statistics; paired t statistic; 95% interval; Pearson correlation; simple regression; Matplotlib.
WORKFLOW	Generate 96 complete participant pairs → validate one session per condition → summarize means and SDs → calculate participant-level differences → estimate the paired effect and interval → calculate correlation and OLS line → export transparent results.
FILES TO OPEN	README.md; analyze.py; performance_sessions.csv; condition_summary.csv; report.json; sleep-reaction-analysis.png; test_analysis.py.

What the results support

Restricted reaction time is 23.9 ms slower on average in the designed sample, with a 20.8–27.0 ms interval. Sleep and reaction time correlate at -0.454; the simple model has $R^2 = 0.206$. These numbers recover effects deliberately encoded in the simulation.

Assumptions and limits

The simple regression does not account for repeated observations, learning, order or carryover. A real study would need a preregistered protocol, verified instruments, missing-data rules and likely a mixed-effects model.

How to describe it to HR or a recruiter

"I built a research analysis that uses the same participants in two conditions, so I compare paired differences rather than treating the sessions as unrelated groups."

How to describe it to a technical interviewer

"I validate complete pairs first, compute the confidence interval from participant-level differences, and separate the paired condition comparison from the across-session correlation and simple regression. I also explain why correlation does not prove causation."

Joins, missing data and validation

LEFT JOIN vs INNER JOIN

INNER JOIN keeps only matched rows from both sides. LEFT JOIN keeps every row from the left table and adds nulls when no right-side match exists. Use LEFT JOIN when unmatched base records are meaningful and must remain visible. Always check the intended grain before and after the join.

Handling nulls

First ask what null means: missing, not applicable, not yet collected or failed match. Do not automatically turn it into zero. Measure missingness, retain raw evidence and choose deletion or imputation only when the purpose and assumptions justify it.

Handling outliers

Validate units and source errors before changing a value. Keep legitimate extremes unless a documented rule says otherwise. Report sensitivity with and without influential records when the decision could change.

Validation checklist

- Confirm primary keys and expected row grain.
- Reconcile row counts through every filter, join and exclusion.
- Check numerator, denominator, period and population for every rate.
- Test duplicates, missing keys, zero denominators and boundary values.
- Compare headline totals with an independent calculation.

KPIs and statistical interpretation

KPI selection

Choose a small set tied to the decision. Pair speed or volume with quality, service and resource use so one metric cannot look better by shifting harm elsewhere. Define every KPI's population, period, formula and owner.

Correlation vs causation

Correlation describes how two variables move together. Causation requires a defensible design that addresses alternatives such as confounding, reverse direction, selection and measurement error. Regression adjustment helps only for measured, correctly modeled factors.

Confidence interval

An interval shows the range of effect sizes reasonably compatible with the model and sample. It is more informative than a pass/fail p-value because it shows magnitude and precision.

Hypothesis test

A p-value is the probability of data at least this incompatible with the null model, assuming the null and test assumptions hold. It is not the probability that the null is true, and statistical significance is not business importance.

Regression

A coefficient is the expected outcome change for a one-unit predictor change under the model. R^2 is the share of outcome variation described in-sample. Neither proves causality or good out-of-sample prediction.

Own the work by being able to change it

The strongest proof is not memorizing the result. It is making a small change, predicting the consequence and verifying the output.

Practice tasks

- Excel: change one valid shift record while preserving on-time + late = due. Confirm the dashboard changes and then restore it.
- SQL: filter one channel and explain whether the filter belongs before or after aggregation.
- Workforce: add one missing engagement score and predict which counts and means should change.
- Research: remove one participant pair and explain why both sessions must be removed for a complete-case paired test.

A clear answer when asked about AI

AI assistance was used to help develop and review these portfolio materials. I can explain the definitions, calculations, code flow, checks and limitations, and I am practicing changes so I can defend the work rather than presenting outputs I do not understand.

Before an interview

- Know one business question, one calculation, one finding and one limitation per project.
- Do not claim a native Power BI report; describe the supplied measures and handoff accurately.
- Do not call contribution profit, association causation or synthetic data real experience.
- If you do not know, explain how you would check rather than guessing.

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Clear definitions. Traceable calculations. Honest limits.